

* Couple the Grh theory to $N=1$ chiral mult.
in $D=4$.

$(\text{Conf}) \oplus (N=1 \text{ chiral}) \equiv \text{Compensating Multiplet}$

Superconf. gauge mult. coupled to Chiral Mult.

$\Downarrow$ Gauge fix

Pure Supergravity

SuperConf. Alg.:

- The Superconf. has the conf. algebra $SO(0,2)$ as
the subalgebra.

Since the super alg. is generated by matrix
elements that are both Bosonic $\oplus$ fermionic, we
find, a general matrix is of the form:

$$\begin{pmatrix} \text{Conf. alg.} & Q, S \\ Q, S & R\text{-symm} \end{pmatrix}$$

For $N=2$, $D=4$ the relevant superalgebra is
is $SU(2,2|1)$

- Describes^{main part of} the Bosonic subalgebra $SU(2,2)$
for which $SO(4,2) \simeq SU(2,2)$

Recall: $M^T M = M$ so, in this case
 $M = \text{diag}(-1, 1, \dots, 1, -1)$

- Refers to the number of Susy generators

* Bosonic Generators: $N=1$ has an R-symmetry group $U(1)_R$ so the full sym group is

$$(\text{Bosonic subalg.}) = SU(2,2) \times U(1)_R$$

Appears in the block diagonal

* $SU(2,2)$ is the covering space of $SO(4,2)$
i.e. $SO(4,2) \cong SU(2,2)/\mathbb{Z}_2$

As in the simple case of $SO(1,3) \cong SU(2,2)/\mathbb{Z}_2$,
this means it contains the same algebra but
allows for fermionic representations

$$\{P_\mu, M_{\mu\nu}, K_\mu, D\} \subset SU(2,2)$$

Aside: (R-symmetry)

$$Q_\alpha \mapsto e^{i\theta \gamma_5} Q_\alpha$$

which has the generator T_R . R-sym is

a $U(1)$ chiral symmetry with

$$[T_R, Q] = -i(\gamma_5)_\alpha^\beta Q_\beta \quad (\text{proof below})$$

$$[T_R, (\text{rest of alg})] = 0$$

So, R-sym is an automorphism of the SUSY

Proof:

$$D=4: T_A: \text{ s.t. } \exists (U_T)_i^j \text{ s.t.}$$

$$[T_A, Q_\alpha] = (U_T)_\alpha^i Q_i \quad \text{[Related by charge]}$$

$$[T_A, Q_\alpha^i] = (U_A)^i_j Q_\alpha^j \quad \text{and} \quad [T_A, Q_\beta^i] = (U_A)^i_j Q_\beta^j$$

$$\therefore (U_A)^i_j = -((U_A)^j_i)^*$$

The R-symmetry generator Q must satisfy the Super-Jacobi identities:

$$[[T_A, T_B], Q] = [T_A, [T_B, Q]] - [T_B, [T_A, Q]]$$

$$\Leftrightarrow f_{AB}^C (U_C) = [U_A, U_B]$$

These are the structure constants we are interested in. Likewise

$$0 = [T_A, \{Q_\alpha^i, Q_\beta^j\}] = \{[T_A, Q_\alpha^i], Q_\beta^j\} + \{[T_A, Q_\beta^j], Q_\alpha^i\}$$

Why does the RHS vanish? By construction

$$[T_A, (\text{Bosonic})] = 0 \quad \text{and} \quad \{Q, Q\} \sim P_a \quad \text{so}$$

$$[T_A, P_a] = 0$$

$$\Rightarrow \{(U_A)^k_i Q_{\alpha k}, Q_\beta^j\} + \{(U_A)^j_k Q_\beta^k, Q_\alpha^i\} = 0$$

$$\Rightarrow (U_A)^k_i \delta_k^j + (U_A)^j_k \delta_k^i = 0$$

$$\therefore (U_A)^i_j = -((U_A)^j_i)^*$$

i.e. $U_A^\dagger = -U_A$, which are anti-hermitian.

→ Anti-Hermitian matrices generate $U(N)$

$$(\text{R-sym group}) \cong U(N)$$

We can normalize: $U_1^2 = -U_2^2 = i U$

generator T_R . Then, given a Majorana

generator for $N=2$ $Q = P_L Q + P_R Q \rightarrow Q + Q^\dagger$

$$[T_R, Q] = -i P_L Q + i P_R Q = -i \gamma_4 Q \quad \checkmark$$

dim	Solnor type	R-symm
$D = 10$	MW	$SO(N_L) \times SO(N_R)$
$D = 9$	M	$SO(N)$
$D = 8, 4$	M	$U(N)$
$D = 7, 5$	S.M	$USp(N)$
$D = 6$	SW	$USp(N_L) \times USp(N_R)$

MW: Maj. Weyl

SM: Symplectic Maj

M: Maj.

SW: Symplectic Weyl

* For $N=1$ chiral: R-symmetry is present only when the theory has a Potential $W(\Phi, \bar{\Phi})$ which is of R-charge 2.

Otherwise the R-symmetry is not existent.
 $\rightarrow$ Here, T_R will act differently w/ the added Carl. generators to the Alg.

Why? : Because, to obtain a theory of

Supergravity, i.e. gravity coupled to the Gravitino, we must Gauge the Symmetries $\Rightarrow$ T_R becomes Gauged (or LOCAL).

$\rightarrow$ (Gauge T_R at the end of our construction)

The Calm before the storm:

-group extension w/ Generators Q_a .

Now, since we have New generators K_a & D must also close the extended SUSY alg. This is done by introducing the:

S_α : (Conformal Supercharge)

- S_α 'squares' to a SCT K_a .
- Both S_α and K_a are to be gauged in the construction.
- ALL Spinors we consider are Maj.

$$[M_{ab}, Q] = -\frac{1}{2} \gamma_{ab} Q, \quad [M_{ab}, S] = -\frac{1}{2} \gamma_{ab} S$$

→ S_α are Lorentz spinors.

* Dilations, which define our conf. weight, give a 5-grading of the Superconf. alg.

h	Gen.
1	P_a
$\frac{1}{2}$	Q
0	D, M_{ab}, T_R
$-\frac{1}{2}$	S
-1	K_a

Not sure, thought -1 must mean $D = -x^\mu \partial_\mu$

Since,

$$[Q, Q] = \frac{1}{2} \gamma^\mu \partial_\mu, \quad [Q, P] = \frac{1}{2} P$$

- For R-symm, we saw that T rotates the susy generators: $[T_R, Q] = -i \gamma_* Q$

Note we used the Normalization $(U_A)' = i$

→ Here we use $(U_A)' = \frac{3i}{2}$

$$\therefore [T_R, Q] = -(\beta i/2) \gamma_* Q$$

$$[T_R, S] = (\beta i/2) \gamma_* S$$

⇒ $P_L Q$ and $P_L S$ have opposite $U(1)_R$ charge.

Since K_a lowers the weight by -1 then

$$[K_a, Q] = \gamma_a S$$

$$[P_a, S] = \gamma_a Q$$

• Use $\bar{Q}^\alpha = Q^\alpha = C^{\alpha\beta} Q_\beta$

$$\{Q_\alpha, Q^\beta\} = -\frac{i}{2} (\gamma^a)_\alpha^\beta P_a$$

$$\{S_\alpha, S^\beta\} = -\frac{i}{2} (\gamma^a)_\alpha^\beta K_a$$

$$\{Q_\alpha, S^\beta\} = -\frac{i}{2} \delta_\alpha^\beta D - \frac{i}{4} (\gamma^{ab})_\alpha^\beta M_{ab} + \frac{i}{2} (\gamma_*)_\alpha^\beta T_R$$

$$\bullet \{[D, S_1], S_2\} + [S_1, S_2], D\} = \{[S_2, D], S_1\} - \frac{i}{2} \{S_1, S_2\} = [D, \{S_1, S_2\}] - \frac{i}{2} \{S_2, S_1\} = 0$$

$$[D, \{S_1, S_2\}] = -\{S_2, S_2\}$$

$$\Leftrightarrow \{S_1, S_2\} \propto K_a$$

• We find the factor by looking at other combinations of the Jacobi.

The Storm...

Gauge fields, transformations, & Curvature:

(i) Assign Gauge fields to all generators of the subalgebra.

P_a	M_{ab}	D	K_a	T	Q_a	S
ξ^a	λ^{ab}	λ_D	λ^a_K	λ_T	ϵ_a	$\mathcal{M}$
e^a_μ	ω^{ab}_μ	b_μ	f^a_μ	A_μ	$\psi_{a\mu}$	ϕ_μ

$$\delta = e^A T_A$$

$$\text{and } D_\mu = \partial_\mu - \delta(B_\mu)$$

Gravitino!

We split P^a into gets and the rest by

$$\delta = e^A T_A = \frac{1}{2} \lambda^{[ab]} M_{cb} + \lambda_D D + \lambda^a_K K_a + \lambda_T T + \epsilon^a Q_a + \mathcal{M} S_a$$

The last two terms are $\bar{E}Q + \bar{\mathcal{M}}S$

* For gets we have $\xi^a = e^a_\mu \xi^\mu$ with

$$\delta(e) B^A_\mu := \partial_\mu e^A + e^C B^B_\mu f^A_{BC}$$

* In the following we underline the terms in which $B^A_\mu \equiv e^a_\mu$, i.e. translations

$$\delta e^a_\mu = -\lambda^a_b e^b_\mu - \lambda_D e^a_\mu + \frac{1}{2} \bar{E} \gamma^a \psi_\mu$$

$$\delta \omega^{ab}_\mu = 2 \lambda^{[c}_\mu \omega^{ab]}_\mu - 4 \lambda^c_K e^a_\mu \lambda^{b]}_c + \frac{1}{2} \bar{E} \gamma^{ab} \phi_\mu + \frac{1}{2} \bar{\mathcal{M}} \gamma^{ab} \psi_\mu$$

$$\delta b_\mu = \partial_\mu \lambda_D + 2 \lambda^c_K e_{\mu c} + \frac{1}{2} \bar{E} \phi_\mu - \frac{1}{2} \bar{\mathcal{M}} \psi_\mu$$

$$\delta f^a_\mu = -\lambda^a_b f^b_\mu + \partial_\mu \lambda^a_K - b_\mu \lambda^a_K + \omega^{ab}_\mu \lambda_{ab} + \lambda_D f^a_\mu + \frac{1}{2} \bar{\mathcal{M}} \gamma^a \phi_\mu$$

$$\delta A_\mu = \partial_\mu \lambda_T + \lambda^a_K A_\mu + \frac{1}{2} \bar{E} \gamma_\mu \psi$$

$$\delta\psi_\mu = -\frac{1}{4}\lambda^{ab}\gamma_{ab}\psi_\mu + (\partial_\mu + \frac{1}{2}b_\mu + \frac{1}{4}\omega_\mu^{ab}\gamma_{ab} - \frac{3}{2}iA_\mu\gamma_5)\epsilon - \gamma_a e_\mu^a \mathcal{N}$$

$$- (\frac{1}{2})\lambda_0\psi_\mu + \frac{3}{2}i\lambda_T\gamma_5\psi_\mu$$

$$\delta\phi_\mu = -\frac{1}{4}\lambda^{ab}\gamma_{ab}\phi_\mu + (\partial_\mu - \frac{1}{2}b_\mu + \frac{1}{4}\omega_\mu^{ab}\gamma_{ab} + \frac{3}{2}iA_\mu\gamma_5)\mathcal{N}$$

$$- \gamma_a f_\mu^a + \frac{1}{2}\lambda_0\phi_\mu - \frac{3}{2}i\lambda_T\gamma_5\phi_\mu + \lambda_K^a\gamma_a\psi_\mu$$

Proof: (i) $\delta b_\mu = \partial_\mu \epsilon^0 + \epsilon^c B_\mu^B f_{Bc}^0$

Note, the only " ϵ^0 " term is given

by $\epsilon^0 = \lambda_0$ and λ_T . We only have

$$\partial_\mu \lambda_0 + \epsilon^c B_\mu^B f_{Bc}^{(0)}$$

so B & C can only be related to objects

$$[T_1, T_2] \sim f_{12}^{(0)} D$$

$$\text{i.e. } [P_c, K_0] = 2(M_{ab} D + M_{ab})$$

$$= f_{P_c K_0}^{(0)} D$$

$$\partial_\mu \lambda_0 + \lambda_K^a e_\mu^a (2M_{ab}) + \dots$$

$$= \partial_\mu \lambda_0 + 2\lambda_K^a e_{\mu a} + \dots$$

• The other terms come from

$$\{Q_\alpha, S^B\} = -\frac{1}{2}\delta_\alpha^B D + \dots$$

$$\therefore \partial_\mu \lambda_0 + 2\lambda_K^a e_{\mu a} + \bar{\epsilon}\phi_\mu(\frac{1}{2}) - \bar{\psi}\psi_\mu(\frac{1}{2})$$

$\rightarrow \pm$ sign occurs from lowering S^B w/
 $C^{\alpha\beta}$. □

(ii) Proof of δe_μ^a

$$\Rightarrow \partial_\mu \xi^a + \partial_\mu \lambda_\mu^a = 0$$

however, in the present case these are

(?) zero. Not sure about

May be from Killing (cont.) eq.
 $\partial_a \xi^b = (M_{ab} \partial_c \xi^c) \frac{1}{D}$, if I had
 to guess

$$e^c \beta_\mu^a f_{bc}^\mu + e^c \beta_\mu^b f_{bc}^\mu = 0$$

$$P_c: [P_a, M_{bc}] = 2\eta_{ab} P_c$$

$$[D, P_c] = P_c, \quad \{Q_\alpha, Q_\beta\} = +\frac{1}{2}(\gamma^a)_{\alpha\beta} P_a$$

$$2\lambda_{ab} e_\mu^a e_\nu^b + \lambda_P e_\mu^a + e^\alpha \psi^\beta (\gamma^a)_{\alpha\beta} \left(\frac{1}{2}\right)$$

(iii) b_μ term in $\delta\psi_\mu$

$$\omega / [D, Q_\alpha] = \frac{1}{2} \delta_\alpha^\beta Q_\beta = Q_\alpha$$

$$\therefore \delta\psi_\mu^\alpha = Q_\mu e^\alpha + e^\kappa \beta_\mu^b f_{bc}^\alpha$$

We want $b_\mu = \beta_\mu^B$ so f_e^α is all

which is left i.e. $\frac{1}{2} \delta_\beta^\alpha = f_\beta^\alpha$

$$\therefore \delta\psi_\mu^\alpha \sim \frac{1}{2} \delta_\beta^\alpha \bar{E}^\beta b_\mu = \frac{1}{2} \bar{E}^\alpha b_\mu \quad \checkmark$$

where $E^B = C^{\beta\alpha} E_\alpha$.

Note the index is up so it really
 is the transformation for $\delta\bar{\psi}_\mu^\alpha$

Lowering w/ $C_{\alpha\beta}$ gives $\delta\psi_{\mu\alpha}$.

* Recall the Curvature:

$$R_{\mu\nu}^A = 2[\partial_\mu \beta_\nu^A] + g_\mu^c \beta_\nu^b f_{bc}^A$$

$$R_{\mu\nu}(P^a) = 2(\partial_\mu + b_{\mu\nu})e_\nu^a + 2\omega_{\mu\nu}^{ab}e_{\nu c} - \frac{i}{2}\bar{\psi}_\mu \gamma^{ac} f_{\nu c}$$

$$R_{mn}(M^{ab}) = 2\partial_{[m}\omega_{n]}^{ab} + 2\omega_{[m}^a\omega_{n]}^b + 8f_{[m}^{[a}e_{n]}^b = \bar{\psi}_m\gamma^a\psi_n$$

...

Constraints:

- We will place constraints to determine the Gauge field S-susy ϕ_m : The remaining independent fields in the Weyl mult.

$$e_m^a, b_m, A_m, \psi_m$$

$$(i) \quad R_{mn}(P^a) = 0$$

$$(ii) \quad e_b^\mu \hat{R}_{\mu\nu}(M^{ab}) = \tilde{\partial} \tilde{R}_\mu{}^a(T)$$

$$(iii) \quad \gamma^\mu R_{\mu\nu}(Q) = 0$$

→ These constraints are invariant under Q-susy.

- A change in constraint redefines the field that we solve for.

Why the above?:

$$\text{For } R_{\mu\nu}(Q) = (-1 - 2\gamma_{\mu\nu})$$

The γ -trace of $R_{\mu\nu}(Q)$ has the same # of components as the field ϕ_m , which we want to solve for, so $\exists!$ a solⁿ of the constraint:

$$(f_{\mu})^{\alpha}_{\beta} K_{\mu\nu}(Q_{\alpha}) \sim (f_{\mu})^{\alpha}_{\beta}$$

which is what is meant since $\phi_{\mu} = (f_{\mu})^{\alpha}_{\beta}$

- These constraints generalize the second order formalism in gravity.

Second order formalism: ω_{μ}^{ab} is defined as a f^{μ} of e^{μ}_a i.e. $\omega_{\mu}^{ab} \in \omega_{\mu}^{ab}(e)$ so it is not an independent field

Similarly, ω_{μ}^{ab} , f_{μ}^a , ϕ_{μ} are all composite fields i.e. they are f^{μ} s of the 'fundamental' gauge fields in the theory: $e^{\mu}_a, b_{\mu}, A_{\mu}, \psi_{\mu}$

Q: Why do we need these constraints?

→ Before we impose the constraints we have an alg. which is closed on all gauge fields of the Weyl mult. However, translations do not act as they should

• See this as $[\delta(\epsilon_1)_{\text{susy}}, \delta(\epsilon_2)_{\text{susy}}] \psi_{\mu} = 0$

- Gauge translations should be replaced by covariant gct's, which is obtained for the frame field by imposing $R_{\mu\nu}(P^a) = 0$

→ The other fields transform as covariant gct's after all these constraints are imposed.

$R_{\mu\nu}(P^a)$: of SUSY Cont. is similar to $R_{\mu\nu}(P^a)$

of the superparticle except we have an extra b_m term

- This b_m can be interpreted as an added term in ω_m^{ab} i.e. Recall be absorbed this for pure Conf. gravity:

$$\omega_m^{ab}(e, b) = \omega_m^{ab}(e) + 2 e_m^{[a} e^{b]v} b_v$$

(i) First Constraint: $R_{\mu\nu}(\rho^a)$

$$\Rightarrow 2(\partial_\mu + b_\mu) e_\nu^a + 2\omega_\mu^{ab} e_\nu^b - \frac{1}{2} \bar{\psi}_\mu \gamma^a \psi_\nu = 0$$

$\Rightarrow$

$$\psi_\mu^{ab}(e, b, \psi) = \omega_\mu^{ab}(e, b) + \frac{1}{2} \bar{\psi}_\mu \gamma^{[a} \psi^{b]} + \underbrace{\frac{1}{4} \bar{\psi}^a \gamma_\mu \psi^b}_{\text{Gravitino torsion}}$$

Gravitino torsion
 T_μ^{ab}

- Symbolically rep. the constraint as

$$\boxed{C(\omega, \psi) = 0} \quad \omega: \text{dependent}, \psi: \text{independent}$$

$\rightarrow$ Assume the constraints are solvable i.e. $\delta C / \delta \omega$ is invertible.

A solution implies that $\exists$ a f^μ $\omega(\psi)$

$$C(\omega(\psi), \psi) = 0 \quad \forall \psi$$

$$\Rightarrow \boxed{\frac{\partial C}{\partial \omega} \frac{\partial \omega}{\partial \psi} + \frac{\partial C}{\partial \psi} = 0}$$

Since $\boxed{\delta \omega = \frac{\partial \omega}{\partial \psi} \delta \psi} \Leftrightarrow \delta C \stackrel{!}{=} 0$,

which is equivalent to δ_ψ

{ For the spin connection ω_m^{ab} we compute $\delta\omega$ by using $\omega(e,b)$ and inserting the transformations of the md. fields e, b, ψ }

* A second way to compute the transformations
 $\rightarrow$ we split the transf. of the dep. fields into the sum of their "old" transformations, dictated by their gauge theory alg. plus extra transf.

$$\delta\omega = (\delta_{\text{gauge}})\omega + \delta_M\omega$$

where the latter is inspired by extended cov. der. $\delta(e) e^A_\mu = \partial_\mu e^A + e^c B^A_c f^A_{bc} + e^B M^A_{B\mu}$



$$\therefore \delta C = \delta_{\text{gauge}} C + \delta_M C = \delta_{\text{gauge}} C + \frac{\partial C}{\partial \omega} \delta_M \omega = 0$$



• First is easy to calc. since the const. is written in terms of curvature:

$$\delta(e) R^A_{\mu\nu} = e^c R^B_{\mu\nu} f^A_{bc}$$

and the invertability of $\partial C / \partial \omega$ allows us then to obtain the new parts of the trans. of the dep. fields

* Let us apply this to the first Constant:

$$(i) R_{\mu\nu}(P^a) = 0$$

$$\delta_{\text{gauge}} R_{\mu\nu}(P) = \bar{E}^{\alpha} R_{\mu\nu} + \text{B.C.}$$

$$\text{we can have } f_{bc}^a \sim 2\eta_{ab} [P, M]$$

$$[D, P_a] \sim f_{(D) a}^P 1$$

$$\{Q_a, Q^b\} = -\frac{1}{2}(\gamma^a)^\alpha_\beta \quad f_{\alpha\beta}^a = \frac{1}{2}\gamma^a_{\alpha\beta}$$

However cannot be $[P, M]$ or $[D, P_a]$

Since $R_{\mu\nu}(P^a) = 0$ thus only $f_{\alpha\beta}^a$ is the only term

$$\therefore \delta_{\text{gauge}} R_{\mu\nu}(P^a) = \bar{E}^\alpha R_{\mu\nu}(Q_\beta) \left(\frac{1}{2}(\gamma^a)^\beta_\alpha\right)$$

$$\delta_{\text{gauge}} R_{\mu\nu}(P^a) = \frac{1}{2} \bar{E}^\alpha \gamma^a_{\alpha\beta} R_{\mu\nu}(Q^\beta)$$

From this, we plug into $\star$

$$0 = \frac{1}{2} \bar{E}^\alpha \gamma^a_{\alpha\beta} R_{\mu\nu}(Q^\beta) + \left(\frac{\partial R_{\mu\nu}(P^a)}{\partial \omega_p^{bc}}\right) \delta_H \omega_p^{ab}$$

• How do we solve for this? We use the trick: Permute the indices three times and add these cyclically:

ex: $T_{\mu\nu\rho}$:

$$[T_{\mu\nu\rho} - T_{\nu\rho\mu} + T_{\rho\mu\nu}]$$

$$R_{\mu\nu}(P^a) \sim 2\omega_{\mu}^{ab} e_{\nu b} = \omega_{\mu}^{ab} e_{\nu b} - \omega_{\nu}^{ab} e_{\mu b}$$

$$\frac{\partial R_{\mu\nu}(P^a)}{\partial \omega_p^{bc}} = (\delta_{\mu}^p \delta_b^a \delta_c^d e_{\nu d} - \delta_{\nu}^p \delta_b^a \delta_c^d e_{\mu d})$$

$$\times \delta_H \omega_p^{bc}$$

$$\Rightarrow \delta_H \omega_{\mu}^{ab} e_{\nu b} - \delta_H \omega_{\nu}^{ab} e_{\mu b}$$

$$\therefore \frac{1}{2} \bar{E} \gamma^a R_{\mu\nu}(\omega) + 2 \delta_{\mu}^{\lambda} \omega_{\lambda\nu}^{ab} e_{\nu}^b = 0$$

We must lower the a index w/ η_{ad}

$$\frac{1}{2} \bar{E} \gamma_a R_{\mu\nu}(\omega) + 2 \delta_{\mu}^{\lambda} \omega_{\lambda\nu}^{\hat{a}b} e_{\nu}^b = 0$$

$$\cancel{\frac{1}{2} \bar{E} \gamma_{\mu} R_{\nu\rho}(\omega)} + 2 \delta_{\mu}^{\lambda} \omega_{\nu\lambda}^{\hat{b}} e_{\rho}^b = 0$$

Notation $\delta_{\mu} \omega \rightarrow \omega$

$$\frac{1}{2} \bar{E} \gamma_{\mu} R_{\nu\rho} + 2 \omega_{\nu\lambda}^{\hat{b}} e_{\rho}^b \quad (a)$$

$$-\frac{1}{2} \bar{E} \gamma_{\nu} R_{\rho\mu} - 2 \omega_{\mu\lambda}^{\hat{b}} e_{\rho}^b \quad (b)$$

$$+\frac{1}{2} \bar{E} \gamma_{\rho} R_{\mu\nu} + 2 \omega_{\rho\lambda}^{\hat{b}} e_{\mu}^b \quad (c)$$

$$\omega_{\nu\mu}^{\hat{b}} e_{\rho}^b - \omega_{\rho\mu}^{\hat{b}} e_{\nu}^b$$

$$\text{note } \omega_{\nu\mu}^{\lambda} e_{\lambda}^b \eta_{bd} e_{\rho}^d = \omega_{\nu\mu}^{\lambda} g_{\lambda\rho}$$

$$(\omega_{\nu\mu\rho} - \omega_{\rho\mu\nu})$$

$$= \omega_{\mu\rho\nu} + \omega_{\nu\rho\mu} = +\omega_{\mu\nu\rho} - \omega_{\nu\rho\mu}$$

$$\omega_{\rho\nu\mu} - \omega_{\mu\rho\nu}$$

$$\therefore 2\omega_{\rho\nu\mu} + \frac{1}{2} \bar{E} (\gamma_{\mu} R_{\nu\rho} - \gamma_{\nu} R_{\rho\mu} + \gamma_{\rho} R_{\mu\nu}) = 0$$

Re-label $\rho \leftrightarrow \mu$

$$2\omega_{\mu\rho\nu} + \frac{1}{2} \bar{E} (\gamma_{\mu} R_{\rho\nu} - \gamma_{\nu} R_{\mu\rho} + \gamma_{\rho} R_{\nu\mu}) = 0$$

$$g^{\nu\lambda} g^{\rho\sigma} () + \frac{1}{2} \bar{E} (\gamma_{\mu} R^{\sigma\lambda} - \gamma^{\lambda} R_{\mu}^{\sigma} + \gamma^{\sigma} R^{\lambda}_{\mu}) = 0$$

$$2 e_a^{\lambda} e_b^{\sigma} \omega_{\mu}^{ab} + \frac{1}{2} \bar{E} (e_a^{\lambda} e_b^{\sigma} \gamma_{\mu} R^{ba} - e_a^{\lambda} e_b^{\sigma} \gamma^{\lambda} R_{\mu}^b + e_b^{\sigma} e_a^{\lambda} (\gamma^b R_{\mu}^a$$

$$2\omega_{\mu}^{ab} + \frac{1}{2} \bar{E} (\gamma_{\mu} R^{ba} - \gamma^a R_{\mu}^b + \gamma^b R_{\mu}^a)$$

$$\Rightarrow 2\omega_m^{ab} = \frac{1}{2} \bar{\epsilon} (\gamma_m R^{ab} + \gamma^a R_m^b - \gamma^b R_m^a)$$

* Extra minus sign missing

$$\delta_\mu \omega_m^{ab} = -\frac{1}{4} \bar{\epsilon} (\gamma_\mu R_m^{ab}) - \frac{1}{2} \bar{\epsilon} \gamma^a R_m^b$$

Thus, we must have: $\delta_{\text{gauge}} \omega = \frac{1}{2} \bar{\epsilon} \gamma^{ab} \phi_\mu$

why? $\Rightarrow$ The formalism is such that we only write the term in $\delta \omega_m^{ab} \propto \bar{\epsilon}$

$$\delta_Q(\epsilon) \omega_m^{ab}(e, b, \phi) = \frac{1}{2} \bar{\epsilon} \gamma^{ab} \phi_\mu - \frac{1}{2} \bar{\epsilon} \gamma^a R_m^b(Q) - \frac{1}{4} \bar{\epsilon} \gamma_\mu R^{ab}(Q)$$

* We can use the third constraint $\gamma_{[\mu} R_{\nu\rho]} = 0$ to simplify this expression

$$\delta_Q(\epsilon) \omega_m^{ab}(e, b, \phi) = \frac{1}{2} \bar{\epsilon} \gamma^{ab} \phi_\mu - \frac{1}{2} \bar{\epsilon} \gamma_\mu R^{ab}(Q)$$

* We define the modified curvature:

$$\hat{R}_{\mu\nu}^A := (r_{\mu\nu}^A - 2 B_{[\mu}^B M_{\nu]}^A)$$

$$\hat{R}_{\mu\nu}(\omega^a) = \hat{R}_{\mu\nu}^{ab} + 8 f_{[a}^c e_{\nu]}^b$$

$$\hat{R}_{\mu\nu}^{ab} = r_{\mu\nu}^{ab} + \bar{\psi}_{[\mu} \gamma^{[c} R_{\nu]}^b(Q) + \frac{1}{2} \bar{\psi}_{[\mu} \gamma_{\nu]} R^{ab}(Q)$$

$$r_{\mu\nu}^{ab} = 2 \partial_{[\mu} \omega_{\nu]}^{ab} + 2 \omega_{[\mu}^a c \omega_{\nu]}^{cb} - \bar{\psi}_{[\mu} \gamma^{ab} \phi_{\nu]}$$

* Solving the second & third constraint:

$$\text{or } \gamma_{[m} R_{\nu p]} = 0$$

$$(2): e^\nu_b \hat{R}_{\mu\nu}{}^{ab} + 8 e^\nu_b f_{[m} e^\mu_{\nu]} = 0$$

$$\hat{R}_\mu{}^a + 2 e^\nu_b (f_{[m} e^\mu_{\nu]} - f_{[m} e^\mu_{\nu]}) = 0$$

$$\Rightarrow \hat{R}_\mu{}^a + 2 e^\nu_b (f_m^a e_\nu^b - f_\nu^a e_m^b - f_m^b e_\nu^a + f_\nu^b e_m^a) = 0$$

$$\hat{R}_\mu{}^a + 2 (f_m^a D - f_m^a - f_\mu^a + f_b^a e_m^b) = 0$$

Solving for f_b^a :

$$\hat{R} + 2(fD - f - f + fD) = 0$$

$$\hat{R} + 4f(D - 1)$$

$$f = -\frac{\hat{R}}{4(D-1)}$$

$$\hat{R}_\mu{}^a + 2f_m^a(D-2) = \frac{\hat{R} e_m^a}{(D-1)} = 0$$

$$f_m^a = -\frac{\hat{R}_\mu{}^a}{2(D-2)} + \frac{\hat{R} e_m^a}{2(D-2)(D-1)} \leftarrow \times 2?$$

will also have $+\frac{1}{4} \epsilon \hat{R}_\mu{}^a(T)$

* Not sure why the factors of $\sim D$ are set to $D=4$ above.

$$\Phi_m = -\frac{1}{2} \gamma^\nu R'_{\mu\nu}(Q) + \frac{1}{12} \gamma_\mu \gamma^{ab} R'_{ab}(Q)$$

where

$$R_{\mu\nu}(Q) = R'_{\mu\nu}(Q) - 2 \gamma_{[m} \delta_{\nu]}$$

$$R'_{\mu\nu}(Q) = 2 \left(\partial_{[m} + \frac{1}{2} \gamma_{[m} \delta_{\nu]} - \frac{3}{2} \epsilon A_{[m} \gamma_{\nu]} + \frac{1}{4} \omega_{[m}{}^{ab} \gamma_{\nu]} \right) \gamma_{\mu]}$$

Note: $f_m^\mu = -\frac{1}{2} \hat{R} = -\frac{1}{2} (R(\omega) - \bar{\psi}_c \gamma^{ab} \phi_b)$
 $\gamma^\mu \phi_m = -\frac{1}{6} \gamma^{\mu\nu} R'_{\mu\nu}(Q)$
 $\gamma^{ab} \phi_b = -\frac{1}{6} \gamma^{abc} R_{bc}(Q)$

Super Cont. trans. rules of a chiral mulf.

* Global susy: Q transforms as a Maj. weyl.

→ Previous Q transformations are valid in

(i) $\bar{\epsilon} Q \rightarrow \epsilon(x)$ i.e. $\in f^4$ of the spacetime.

(ii) Derivatives of the field content of chiral mulf

$$\partial_\mu Z \text{ \& } \partial_\mu P_L \chi \rightarrow D_\mu Z, D_\mu P_L \chi$$

Where D_μ is the super covariant derivative

$$D_\mu := \partial_\mu - \bar{P}_\mu^A T_A \leftarrow \text{Generators of Conf(MD)}$$

• First try to understand how Dilatations act in this setting.

⇒ Finding the cond. weight h for each field of chiral mulf.

ex: Z : has c.c. h .

$$[D, Q] = \frac{1}{2} Q$$

$$\Leftrightarrow [D, Q] Z = D Q Z - Q D Z = D(Q Z) - h(Q Z)$$

$$\delta Z = \frac{1}{2} \bar{\epsilon} P_L \chi$$

$$\Rightarrow \delta Z = [EQ, Z] \Rightarrow Q Z = \frac{1}{2} P_L \chi$$

$$\frac{1}{2} [D P_L \chi - h P_L \chi] = \frac{1}{2} (\frac{1}{2} P_L \chi)$$

$$\Leftrightarrow D P_L \chi = (h + \frac{1}{2}) P_L \chi$$

∴ $P_L \chi$ (the susy partner of Z) has cond. weight

$$h' = h + \frac{1}{2}$$

• Q applied to an element of the chiral mulf. raises its cond. weight by $\frac{1}{2}$.

→ Further, note, the weight of $\bar{E} P_L \chi$ must be that of Z , i.e. h . or:

$$DZ = h \Leftrightarrow D(\bar{E} P_L \chi) = h \bar{E} P_L \chi + h_\chi \bar{E} P_L \chi$$

$$\Rightarrow h = h \bar{E} + h_\chi$$

However, the previous exercise has shown that $h_\chi = h + 1/2$
 $\therefore h = h \bar{E} + (h + 1/2) \Leftrightarrow \boxed{h \bar{E} = -1/2}$

* The can. weight of the susy transformation parameters ϵ have can. weight $= -1/2$. Q has $1/2$

→ Naturally, we would find, for supercan.

$$[D, S] = -1/2 S$$

$$S_{s.c.} = \mathcal{M} S \Leftrightarrow \boxed{h_{\mathcal{M}} = 1/2}$$

For a chiral mult. ($D_i Z = 0$) the auxiliary term thus has weight:

$$\delta F = \frac{1}{\delta^2} \bar{E} \delta P_L \chi$$

Recall that $\partial_m Z$ has can. weight $h+1$. So the terms in $\bar{E} P_L \chi$ leave h while $\gamma^m \partial_m$ is $+1$

$$D F = h_F F \Leftrightarrow h_F = h+1$$

$$\left(\begin{array}{l} Z: h \\ P_L \chi: h+1/2 \end{array}, F: h+1 \right)$$

• T_R transformations: Define a chiral weight c s.t.

$$\boxed{\delta_{T_R} Z := i c \Lambda_T Z}$$

and similarly for all other fields.

$$[T, Q] = -\frac{3}{2} i \gamma_\mu Q \quad \& \quad [T, S] = \frac{3}{2} i \gamma_\mu S$$

These imply that $P_L \epsilon$ (or $\bar{E} P_L$) has chiral weight $1/2$

$$[T, Q] Z = T P_L \chi - Q(T Z) = -3/2 i \gamma_\mu (P_L \chi)$$

$$\Rightarrow T(P_L \chi) - Q(i c Z) = -1/2 i \gamma_\mu P_L \chi$$

$$T(P_L \chi) = i c P_L \chi - 3/2 i \gamma_\mu P_L \chi$$

$$= i(c - 1/2 \gamma_r) P_L \chi$$

$$\gamma_r P_L = \frac{1}{2} \gamma_r (1 + \gamma_r) = -P_R$$

$$\Rightarrow = i c P_L \chi + i 1/2 P_R \chi$$

$\Leftrightarrow$ (?? How) $\Rightarrow P_L \psi$ and $\bar{\psi} P_L$ have chiral weight $3/2$ while $P_R \psi$ has chiral weight $-3/2$

Recall, $\delta_T = \lambda_T T$

$$\Rightarrow [T, Q] = -3/2 i \gamma_4 Q$$

$$\text{or } \lambda_T \bar{\psi} [T, Q] \psi = -3/2 i \lambda_T (\bar{\psi} \gamma_4) Q \psi$$

$$[\delta_T, \delta_Q] \psi = \delta_T \delta_Q \psi - \delta_Q \delta_T \psi$$

$$= \delta_T (\frac{1}{2} \bar{\psi} P_L \chi) - \delta_Q (i c \lambda_T \psi)$$

$$= \frac{1}{2} \delta_T (\bar{\psi} P_L \chi) - i c \lambda_T \delta_Q \psi$$

$$\frac{1}{2} \delta_T (\bar{\psi} P_L \chi) - i c \lambda_T \delta_Q \psi = 3/2 i \lambda_T (\gamma_4) \frac{1}{2} \bar{\psi} P_L \chi$$

$$\delta_T (\bar{\psi} P_L \chi) - i c \lambda_T (\bar{\psi} P_L \chi) = 3/2 i \lambda_T \gamma_4 \bar{\psi} P_L \chi$$

$$\delta_T (\bar{\psi} P_L \chi) = i \lambda_T (c \bar{\psi} P_L \chi + 3/2 \gamma_4 \bar{\psi} P_L \chi)$$

$$\hookrightarrow \delta_T (\bar{\psi}) P_L \chi + \bar{\psi} \delta_T (P_L \chi) = \delta_T (\bar{\psi}) P_L \chi + i \lambda_T c \chi \bar{\psi} P_L \chi$$

however $c_\chi = c + 1/2$

$$\therefore i c \lambda_T \bar{\psi} P_L \chi + i \lambda_T (c + 1/2) \bar{\psi} P_L \chi = i c \lambda_T \bar{\psi} P_L \chi + 3/2 i \lambda_T \gamma_4 \bar{\psi} P_L \chi$$

$$(c + 1/2) \bar{\psi} P_L \chi = 3/2 \gamma_4 \bar{\psi} P_L \chi$$

$\rightarrow$ Not too sure about this $3/2$

$$\left\{ \begin{array}{ll} P_L \psi \text{ \& } \bar{\psi} P_L : & c = 3/2 \\ P_R \psi \text{ \& } \bar{\psi} P_R : & c = -3/2 \\ P_L \eta : & c = -3/2 \\ P_R \eta : & c = 3/2 \end{array} \right.$$

These are Chiral Weights and NOT Conformal weights.

Claim: The chiral weights of the fields and the parameters require that $\mathcal{Z}$ is invariant under $\mathcal{S}$ supersymmetry.

Proof: Since η in $\mathcal{S} = \mathcal{H}_S$ has weight $+3/2$ $\mathcal{Z}$

Can only transform into a field of weight $h = 1/2$:

$$[D, S]Z = -\frac{1}{2}SZ$$

$$h_S SZ - h SZ = -\frac{1}{2}SZ \Rightarrow \boxed{h_S = (h - \frac{1}{2})}$$

- Recall, the Covariant derivative $\mathbb{D}_a$ carries a weight 1

$\therefore$ Only possibility is $\boxed{S_Z Z = 0}$

* We can use this fact to relate the chiral weight & the Conformal weight

Start by looking at $[Q, S]Z$

Recall that, for a supersymmetry transf.:

$$\delta(\epsilon_1) \delta(\epsilon_2) = ((\bar{\epsilon}_1)^\alpha Q_\alpha) ((\bar{\epsilon}_2)^\beta Q_\beta)$$

however $(\bar{\epsilon}_2)^\alpha Q_\alpha$ is Bosonic so

$$\delta(\epsilon_1) \delta(\epsilon_2) = (\bar{\epsilon}_2)^\beta (\bar{\epsilon}_1)^\alpha Q_\alpha Q_\beta$$

$$\delta(\epsilon_2) \delta(\epsilon_1) = (\bar{\epsilon}_1)^\beta (\bar{\epsilon}_2)^\alpha Q_\alpha Q_\beta$$

$$\begin{aligned} [\delta(\epsilon_1), \delta(\epsilon_2)] &= (\bar{\epsilon}_2)^\beta (\bar{\epsilon}_1)^\alpha (Q_\alpha Q_\beta) - (\bar{\epsilon}_1)^\beta (\bar{\epsilon}_2)^\alpha Q_\alpha Q_\beta \\ &= (\bar{\epsilon}_2)^\beta (\bar{\epsilon}_1)^\alpha (Q_\alpha Q_\beta - Q_\beta Q_\alpha) \end{aligned}$$

$$\boxed{[\delta(\epsilon_1), \delta(\epsilon_2)] = \bar{\epsilon}_2^\beta \bar{\epsilon}_1^\alpha \{Q_\alpha, Q_\beta\}}$$

$$\{Q_\alpha, Q_\beta\} = -\frac{1}{2}(\gamma^\mu)_{\alpha\beta} P_\mu, \quad \{S_\alpha, S^\beta\} = -\frac{1}{2}(\gamma^\mu)_{\alpha}{}^\beta K_\mu$$

$$\{Q_\alpha, S^\beta\} = -\frac{1}{2}\delta_\alpha^\beta D - \frac{1}{4}(\gamma^{\mu\nu})_{\alpha}{}^\beta M_{\mu\nu} + \frac{1}{2}\epsilon_{\alpha\beta} T_R$$

Require gen's: since $[\delta_a, \delta_b] \sim \delta_{[a,b]}$ then

we know that translations must be changed to gen's:

* We note, a gen. has parameter ξ^μ s.t., for a real scalar field ϕ :

$$\boxed{\delta_{\text{gen}}(\xi) \phi^i = \xi^\mu \partial_\mu \phi^i + (\xi^\mu A_\mu^A) (t_A)^i{}_j \phi^j}$$

* Here we have

$$\xi^\mu = \frac{1}{2} \bar{\epsilon}_2 \gamma^\mu \epsilon_1, \quad \lambda_D = \frac{1}{2} \bar{\epsilon} \epsilon, \quad \lambda_T = -\frac{1}{2} \bar{\epsilon} \gamma_5 \epsilon$$

$$\lambda_\kappa = \frac{1}{2} \bar{\eta}_2 \gamma^\kappa \eta_1, \quad \lambda^{\kappa\alpha} = \frac{1}{4} \bar{\epsilon} \gamma^\kappa \eta$$

(i) Pure Susy term

$$\begin{aligned} (\epsilon_1)_\rho (\epsilon_2)^\alpha \{Q_\alpha, Q^\rho\} &= (\epsilon_1)_\rho (\epsilon_2)^\alpha (-\frac{1}{2} (\gamma^0)_\alpha^\rho P_0) \\ [\delta_\alpha(\epsilon_1), \delta_\alpha(\epsilon_2)] &= (-\frac{1}{2} \bar{\epsilon}_1 \gamma^\alpha \epsilon_2) P_\alpha = (\frac{1}{2} \bar{\epsilon}_2 \gamma^\alpha \epsilon_1) P_\alpha \\ &= \delta_{\text{cgct}}(\xi^\alpha) P_\alpha \end{aligned}$$

(ii) Super conf. term

$$\begin{aligned} (\eta_2)_\rho \bar{\eta}_1^\alpha \{S_\alpha, S^\rho\} &= -\frac{1}{2} (\gamma^\alpha)_\alpha^\rho K_\alpha \Leftrightarrow [\delta_S(\eta_1), \delta_S(\eta_2)] = \delta_\kappa(\lambda^\kappa) K_\kappa \\ \text{with } \lambda_\kappa^\alpha &= \frac{1}{2} \bar{\eta}_2 \gamma^\alpha \eta_1 \end{aligned}$$

(iii) Mixed super term:

$$[\delta_S(\eta_1), \delta_\alpha(\epsilon)] = \delta_D(\lambda_D) + \delta_M(\lambda^{\alpha\beta}) + \delta_T(\lambda_T)$$

• The Second commutator tells us that

$$\boxed{\delta_\kappa z = 0}$$

So the complex scalar in the chiral multiplet transforms trivially under SCT's as well as its supersymmetric counterpart $\delta_S z = 0$.

$$\begin{aligned} \rightarrow [\delta_S(\eta), \delta_\alpha(\epsilon)] z &= \delta_S \delta_\alpha z - \delta_\alpha \delta_S z \\ &= \delta_S (\frac{1}{\sqrt{2}} \bar{\epsilon} P_L \chi) \end{aligned}$$

$$= \frac{1}{\sqrt{2}} \bar{\epsilon} P_L \delta_S(\eta) \chi$$

here $\bar{\epsilon} P_L$ does not appear in the result.

However this should be

$$\frac{1}{\sqrt{2}} \bar{\epsilon} P_L \delta_S(\eta) \chi = \delta_D z + \cancel{\delta_M z} + i \delta_T z$$

since we know $\delta_D z = h z$

$$\Rightarrow \frac{1}{\sqrt{2}} \bar{\epsilon} P_L \delta_S(\eta) \chi = h z + i \delta_T z$$

$$\frac{1}{\sqrt{2}} \bar{\epsilon} P_L \delta_S(\eta) \chi = \lambda_D h z - \lambda_T c z$$

• Split $\bar{\epsilon} = \bar{\epsilon} P_L + \bar{\epsilon} P_R$

$$\frac{1}{\sqrt{2}} (\bar{\epsilon} - \bar{\epsilon} P_R) \delta_S(\eta) \chi = \lambda_D h z - \lambda_T c z$$

$$\frac{1}{\sqrt{2}} \bar{\epsilon} P_L \delta_s(\eta) \chi = \frac{1}{2} (\bar{\epsilon} P_L + \bar{\epsilon} P_R) \eta h z + \frac{1}{2} (\bar{\epsilon} P_L + \bar{\epsilon} P_R) \gamma_* \eta (c z)$$

$$LHS = \frac{1}{2} \bar{\epsilon} P_L (\eta(h z) + \gamma_* \eta(c z)) + \frac{1}{2} \bar{\epsilon} P_R (\eta h z - \eta c z)$$

- No $\bar{\epsilon} P_R$ terms on LHS so we conclude $(\frac{1}{2} \bar{\epsilon} P_R \eta)(h z - c z) = 0 \Rightarrow \boxed{c = h}$

- Consider the left-handed Chiral terms $\checkmark$

$$\frac{1}{\sqrt{2}} \bar{\epsilon} P_L \delta_s(\eta) \chi = \frac{1}{2} h z \bar{\epsilon} P_L \eta + \frac{1}{2} h z \bar{\epsilon} P_L \eta$$

$$\boxed{\delta_s(\eta) P_L \chi = \sqrt{2} z P_L \eta}$$

$$\delta P_L \chi = \frac{1}{\sqrt{2}} P_L (\not{\partial} z + F) \epsilon$$

- * Similarly, we can find the local Super coord. transformations for the auxiliary field F in $(D_a \bar{\epsilon} = 0)$.

$$\begin{aligned} [\delta_s, \delta_a] \chi &= \delta_s \delta_a \chi - \delta_a \delta_s \chi \\ &= \delta_s (\not{\partial} z + F) \epsilon - \delta_a (\sqrt{2} h z \eta) \end{aligned}$$

- * NOTE: Since we are gauging our symmetries, we must replace $\partial_a z \mapsto D_a z$ (should be D_a ?)

$$\begin{aligned} \delta_s (\not{\partial} z + F) \epsilon - \sqrt{2} h \eta \frac{1}{\sqrt{2}} \bar{\epsilon} P_L \chi \\ = (\delta_s F) \epsilon - \sqrt{2} h \eta \bar{\epsilon} P_L \chi + \delta_s (\not{\partial} z) \epsilon \end{aligned}$$

- Ⓐ Lemma states $\delta_s(\not{\partial} z)$ can only depend on the values previously found.

$$\Rightarrow D_a z = \partial_a z - A_a^A T_A z$$

$$\begin{aligned} \delta(D_a z) &= \delta_s(\partial_a z) - (\delta A_a^A) T_A z \\ &= -\delta A_a^A T_A z \quad \text{(scalar)} \quad \downarrow \end{aligned}$$

$$= -\delta_\epsilon^a P_a z - \delta_\epsilon^b \cancel{\omega_{ab}^c} z - \delta_\epsilon^b D_b z - \delta_\epsilon^a Q_a z$$

First term vanishes as scalars are invariant under translations

the first term vanishes since $\delta_5 \psi \sim \chi$ so it is a scalar
 $\delta_5 Z = 0$, we are simply left w/ the last term which
 is a SUSY:

$$\gamma^\mu \delta_5 (\mathbb{D}_\mu Z) = -\gamma^\mu (\gamma_\alpha \epsilon^\alpha \mathbb{M}) \frac{1}{\sqrt{2}} P_L \chi = \gamma^\mu \gamma_\alpha \mathbb{M}$$

$$\begin{aligned} (\gamma_\alpha)_\alpha^\beta \mathbb{M}_\beta &= (\gamma_\alpha)_\alpha^\beta \mathbb{M}^\gamma C_{\beta\gamma} = -(\gamma_\alpha)_\alpha^\beta C_{\beta\gamma} \mathbb{M}^\gamma \\ &= -(\gamma_\alpha)_\alpha^\gamma \mathbb{M}^\gamma = -\bar{\eta}^\gamma (\gamma_\alpha)_\gamma^\alpha \end{aligned}$$

$$-\frac{1}{\sqrt{2}} \gamma^\mu \bar{\eta}^\gamma (\gamma_\alpha)_\gamma^\alpha P_L \chi$$

$$\textcircled{a} = \gamma^\mu (-\frac{1}{\sqrt{2}} \bar{\eta} \gamma_\alpha P_L \chi)$$

$$\therefore [\delta_3, \delta_2] \chi$$

$$= \delta_3 F \in -\sqrt{2} \mathbb{M} \bar{\epsilon} P_L \chi - \frac{1}{\sqrt{2}} \gamma^\mu \bar{\eta} \gamma_\alpha P_L \chi \in$$

$$= \delta_D \chi + \delta_M \chi + \delta_T \chi$$

$$= (h + \frac{1}{2}) (\bar{\epsilon} \mathbb{M}) \chi + \frac{1}{2} \bar{\epsilon} \gamma^\mu \mathbb{M} [-\frac{1}{\sqrt{2}} \gamma_\alpha] \chi - \frac{1}{2} \bar{\epsilon} \gamma_\mu \mathbb{M} (h + \frac{1}{2}) \chi$$

$$\bullet \bar{\epsilon}^\alpha (\gamma^\mu)_\alpha^\beta \mathbb{M}_\beta (\gamma_\alpha)_\gamma^\delta \chi_\delta$$

$$\rightarrow \gamma^\alpha \gamma^\beta \gamma_\alpha = \frac{1}{4} (\gamma^\alpha \gamma^\beta + \gamma^\beta \gamma^\alpha) (\gamma_\alpha \gamma_\beta - \gamma_\beta \gamma_\alpha)$$

$$= \frac{1}{4} (\gamma^\alpha \gamma^\beta \gamma_\alpha \gamma_\beta - \gamma^\beta \gamma^\alpha \gamma_\alpha \gamma_\beta + \gamma^\alpha \gamma^\beta \gamma_\beta \gamma_\alpha + \gamma^\beta \gamma^\alpha \gamma_\beta \gamma_\alpha)$$

$$= \frac{1}{2} (\gamma^\alpha \gamma^\beta \gamma_\alpha \gamma_\beta - \gamma^\alpha \gamma^\beta \gamma_\beta \gamma_\alpha)$$

$$= \frac{1}{2} (-\gamma^\beta \gamma^\alpha \gamma_\alpha \gamma_\beta + 2 \eta^{\alpha\beta} \gamma_\alpha \gamma_\beta - \gamma^\beta \gamma^\alpha \gamma_\beta \gamma_\alpha)$$

$$= (\eta^\alpha \gamma_\alpha - \gamma^\beta \gamma^\alpha \gamma_\alpha \gamma_\beta)$$

$$= (D^2 - 1) \mathbb{1}$$

$$\Rightarrow \bar{\epsilon}^\alpha (D^2 - 1) \delta_\alpha^\beta \delta_\beta^\gamma \chi_\gamma \mathbb{M}_\beta = (\bar{\epsilon} \mathbb{M}) (D^2 - 1) \chi$$

$$\Leftrightarrow -\frac{1}{4} (\bar{\epsilon} \mathbb{M}) (D^2 - 1) \chi$$

$$\text{LHS} = (h + \frac{1}{2}) \frac{1}{2} (\bar{\epsilon} \mathbb{M}) \chi - \frac{1}{4} (\bar{\epsilon} \mathbb{M}) (D^2 - 1) \chi + \frac{1}{2} \bar{\epsilon} \gamma_\mu \mathbb{M} (h + \frac{1}{2}) \chi$$

$$\rightarrow \text{RHS} = (\delta_3 F) \in + \sqrt{2} h \bar{\epsilon} \mathbb{M} P_L \chi - \frac{1}{2} \gamma^\mu \bar{\eta} \gamma_\alpha P_L \chi \in$$

(5)

$$\textcircled{b} = -\frac{1}{2} (\gamma^\alpha)_\alpha^\beta \bar{\eta}^\beta (\gamma_\alpha)_\gamma^\delta (P_L \chi)_\delta \in_\beta$$

$$= -\frac{D}{2} \epsilon_\alpha (\bar{\eta} P_L \chi)$$

$$\text{RHS} = (\delta_3 F) \in + \sqrt{2} h (\bar{\epsilon} \mathbb{M}) P_L \chi - \frac{D}{2} \in (\bar{\eta} P_L \chi)$$

$$\text{LHS} = (h + \frac{1}{2}) \frac{1}{2} (\bar{\epsilon} \mathbb{M}) \chi - \frac{1}{4} (D^2 - 1) (\bar{\epsilon} \mathbb{M}) \chi + \frac{1}{2} \bar{\epsilon} \gamma_\mu \mathbb{M} (h + \frac{1}{2}) \chi$$

$$\begin{aligned} \bar{\epsilon}^\alpha M_\beta \chi_\alpha &\sim (M_\beta)^\alpha_\alpha = \frac{1}{2^n} \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{2} \gamma_{\mu_1 \dots \mu_n} \right) \left(\gamma^{\mu_1 \dots \mu_n} \delta^\alpha_\beta M^\beta_\alpha \right) \quad \left\{ \text{or } n=2n \right\} \\ &= \frac{1}{2^n} \left(M^\delta_\delta \delta^\alpha_\alpha + (\gamma_{\mu_1})^\beta_\alpha (\gamma^{\mu_1})^\alpha_\beta M^\delta_\delta + \frac{1}{2} (\gamma_{\mu_1 \mu_2})^\beta_\alpha (\gamma^{\mu_1 \mu_2})^\alpha_\beta M^\delta_\delta \right. \\ &\quad \left. + \frac{1}{6} (\gamma_{\mu_1 \mu_2 \mu_3})^\beta_\alpha (\gamma^{\mu_1 \mu_2 \mu_3})^\alpha_\beta M^\delta_\delta + \frac{1}{24} (\gamma_{\mu_1 \mu_2 \mu_3 \mu_4})^\beta_\alpha (\gamma^{\mu_1 \mu_2 \mu_3 \mu_4})^\alpha_\beta M^\delta_\delta \right) \\ \gamma^{\mu_1 \dots \mu_n} &= i \epsilon^{\mu_1 \dots \mu_n} \gamma_5, \quad \gamma^{\mu_1 \mu_2 \mu_3} = \gamma^{\mu_1 \mu_2} \gamma^{\mu_3} \end{aligned}$$

Shift happens ... (Expand χ via duals)

$$\Rightarrow \boxed{\delta_S(n)F = \sqrt{2}(1-h) \bar{n} P_L \chi}$$

Complete Superconformal transf. for the chiral Multiplet
(w/o get's ξ Lorentz)

$$\begin{aligned} \delta Z &= (L\lambda_0 + i h \lambda_T) Z + \frac{1}{\sqrt{2}} \bar{\epsilon} P_L \chi + 0 \\ \delta P_L \chi &= [(h + \frac{1}{2})\lambda_0 + (h - \frac{1}{2})i\lambda_T] P_L \chi + \frac{1}{\sqrt{2}} P_L (\not{D}Z + F) \epsilon + \sqrt{2} h Z \not{P} n \\ \delta F &= [(h+1)\lambda_0 + (h-1)i\lambda_T] F + \frac{1}{\sqrt{2}} \not{D} P_L \chi + \sqrt{2}(1-h) \bar{n} P_L \chi \end{aligned}$$

- Dilations (δ_0)
- Super symmetry (δ_α)
- R-sym (δ_{Tn})
- Super Conformal (δ_S)

* For chiral superfields we find that the R-symmetry $U(1)_R$ charge is completely determined by the Conf. weight of the field in the multiplet.
(~DUP to normalization of the U_1 matrix)

* Any supercovariant action will also be T_R -symmetric (even locally)
 $\rightarrow$ Super conf. cov. der.:

$$\boxed{D_a Z = e_a^\mu (\partial_\mu Z - h b_\mu - i A_\mu - \frac{1}{\sqrt{2}} \not{P}_\mu P_L \chi)}$$

Note: there is no Lorentz term as Z is a scalar $P_L \chi$, however, contains Lorentz / Poincare terms, i.e. $\sim \omega_{\mu\nu}^{\alpha\beta}$
Spin connection terms are also present:

$$\boxed{D_a P_L \chi = e_a^\mu P_L \left[\partial_\mu + \frac{1}{4} \omega_{\mu}^{\alpha\beta} \gamma_{\alpha\beta} - (h + \frac{1}{2}) b_\mu - i (h - \frac{1}{2}) A_\mu \right] \chi - \frac{1}{\sqrt{2}} (\not{D}Z + F) \not{P}_\mu \epsilon - \sqrt{2} h Z \not{P}_\mu \epsilon}$$

• In global susy F transforms as a total der. Therefore FF^* is the appropriate 'Lagrange multiplier'

↳ In a theory of gravity we should expect $\sim eF$

Since the measure has changed from $M^D \rightarrow M^D$

• e : Conf. weight = $-4 = w_e$

So F should carry weight $+4$. From above we

see it carries $[h+1] \Leftrightarrow [h=3]$

• For this choice we also find that F becomes invariant under R -symmetry as well

$\int d^4x eF$: NOT invariant under SCT's as $\delta_\alpha F$ & $\delta_\beta F$ lower/rise the weight/field.

The modified integral becomes

(Master Formula)

$$S_F = \int d^4x e \operatorname{Re} \left\{ F + \frac{1}{\sqrt{2}} \bar{\Psi}_\mu \gamma^\mu P_L \chi + \frac{z}{2} \bar{\Psi}_\mu \gamma^{\mu\nu} P_R \Psi_\nu \right\}$$



(Both $\operatorname{Re}\{\cdot\}$ & $\chi_m\{\cdot\}$ give SC. invariant Actions for a chiral multiplet of Conf. weight $h=3$.)

Proof: We prove that S_F is invariant under δ_S :

$$\delta_S S_F = \int d^4x \delta_S (e \operatorname{Re}\{\cdot\})$$

$$= \int d^4x (\delta_S e \operatorname{Re}\{\cdot\} + e \operatorname{Re}\{\delta_S F + \frac{1}{\sqrt{2}} \delta_S (\bar{\Psi}_\mu \gamma^\mu P_L \chi) + \frac{z}{2} \delta_S (\bar{\Psi}_\mu \gamma^{\mu\nu} P_R \Psi_\nu)\})$$

$$\text{v/ } \delta_S e = 0$$

$$\textcircled{a}: \delta_S F = \sqrt{2}(1-h) \bar{\eta} P_L \chi \Rightarrow \operatorname{Re}\{\delta_S F\}$$

$$= \sqrt{2}(1-h) \bar{\eta} P_L \chi \quad \leftarrow$$

$$\textcircled{b} \frac{1}{\sqrt{2}} \delta_S \bar{\Psi}_\mu \gamma^\mu P_L \chi + \frac{1}{\sqrt{2}} \bar{\Psi}_\mu \gamma^\mu P_L (\sqrt{2} h z \eta)$$

$$\hookrightarrow = -\bar{\eta} e^a \gamma_a (\frac{1}{\sqrt{2}} P_L \chi)$$

$$\textcircled{c} \frac{z}{2} (-\bar{\eta} e^a \gamma_a) \gamma^{\mu\nu} P_R \Psi_\nu - \frac{z}{2} \bar{\Psi}_\mu \gamma^{\mu\nu} P_R (\gamma_\nu \eta)$$

$$\hookrightarrow \int d^4x e \operatorname{Re} \left\{ \sqrt{2}(1-h) \bar{\eta} P_L \chi - \frac{1}{2} \bar{\eta} \gamma_\mu \gamma^\mu P_L \chi + h z \bar{\Psi}_\mu \gamma^\mu P_L \eta - \frac{1}{2} z \bar{\eta} \gamma_\mu \gamma^{\mu\nu} P_R \Psi_\nu - \frac{1}{2} z \bar{\Psi}_\mu \gamma^{\mu\nu} P_R (\gamma_\nu \eta) \right\}$$

$$= 0$$

$$-\frac{1}{2} \epsilon \bar{\chi} (D - 1) \gamma^\mu P_R \psi - \frac{1}{2} \epsilon \bar{\psi}_m \gamma^{\mu\nu} P_R \chi_\nu \mathcal{M}$$

$$\left\{ \begin{aligned} \{\chi_\mu, \chi_\nu\} &= 0 \Rightarrow \gamma^\mu P_\pm = \pm \gamma_\mu (1 \pm \gamma_5) = \pm (\gamma_\mu \pm \gamma_\mu \gamma_5) \\ \gamma^\mu P_\pm &= \pm \frac{1}{2} (1 \pm \gamma_5) \gamma_\mu = P_\pm \gamma_\mu \end{aligned} \right.$$

$$\int d^4x \operatorname{Re} \left\{ \sqrt{2} (1-h) \bar{\chi} P_L \chi - D \bar{\chi} P_L \chi + h \epsilon \bar{\psi}_m \gamma^\mu P_L \mathcal{M} \right. \\ \left. - \frac{1}{2} (D-1) \epsilon \bar{\chi} \gamma^\mu P_R \psi - \frac{1}{2} \epsilon \bar{\psi}_m \gamma^\mu (D-1) P_L \mathcal{M} \right\}$$

Terms cancel

$$\int d^4x \operatorname{Re} \left\{ \sqrt{2} (1-h) \bar{\chi} P_L \chi - D \bar{\chi} P_L \chi + h \epsilon \bar{\psi}_m \gamma^\mu P_L \mathcal{M} \right\}$$

$\rightarrow$ Since we are building a $D=4$ dim theory w/ ϵ of Conf. weight $h=3$ we find

$$\int d^4x \operatorname{Re} \left\{ \sqrt{2} (-2) \bar{\chi} P_L \chi + \frac{4}{\sqrt{2}} \bar{\chi} P_L \chi + 3 \epsilon \bar{\psi}_m \gamma^\mu P_L \mathcal{M} \right\}$$

$\sim$ third term cancels (same as w/ $\mathcal{L}$ term) $\square$

* A chiral mult. is determined by its lowest component both in global SUSY and in the superconf setting.

$\Rightarrow$ If one field transforms as $\mathbb{Z}$ w/ χ replaced by some arbitrary spinor, then the superconf. algebra determines that this χ should transform as the χ above for some F .

Superconf. action of the chiral Multiplet:

Idea: Generalize

$$S = \int d^Dx \left(\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \frac{D-2}{8(D-1)} R \phi^2 \right)$$

which is a purely gravitational theory coupled to scalars

• $\phi(x)$ is not physical

$\rightarrow$ Need a scalar field w/ weight $h=1$ (as in S above w/ ϕ)

$\sim$ We can obtain " ϕ " through $\mathbb{Z}$ in the multiplet of $\{\mathbb{Z}, P_L \chi, F\}$ with $[h=1]$.

Steps:

chiral multiplet Φ s.t. $D_\alpha \Phi = 0$ has the field content $\{\bar{\Phi}, P_L \Phi, F\}$

$\rightarrow$ The fields $\{\bar{\Phi} := \bar{\Phi}, P_L \chi := \partial P_R \chi, F' := D \bar{\Phi}\}$ are also components of a chiral mult.

The weyl weight of $\bar{\Phi}$ is $h_{\bar{\Phi}} = 2$ since $h_\chi = 3/2$

(ii) Consider the product multiplet:

$$\{\bar{\Phi} \bar{\Phi}, P_L(\bar{\Phi} \partial \chi + \bar{\chi} F), F\bar{\Phi} + \bar{\chi} D \bar{\Phi} - \bar{\chi} P_L \partial \chi\}$$

This is from $\Phi^i \bar{\Phi}^j \rightarrow D_\alpha (\Phi^i \bar{\Phi}^j) = 0$

where $\bar{\Phi}^i$ of Φ^j is

The lowest weight state is $h=3$ while the highest weight state has $h=4$

(iii) The Action function:

$$S_F = \int d^4x \operatorname{Re} \left\{ \bar{\Phi} + \frac{1}{12} \bar{\chi}_\mu \gamma^\mu P_L \chi + \frac{1}{2} \bar{\chi} \bar{\Phi} \gamma^{\mu\nu} P_R \chi_\nu \right\} e$$

gives a locally superant. mv. action for a chiral mult.

whose weyl weight is $h_F = 4$

Q: why? recall that e carries weyl weight $h_e = -4$

Since $e = \frac{1}{4!} \epsilon^{a_1 \dots a_4} e_{a_1}^{m_1} e_{a_2}^{m_2} e_{a_3}^{m_3} e_{a_4}^{m_4} \epsilon_{m_1 \dots m_4}$ it carries

4 values of e_a^m so (eF) must have weight 0.

$$S_F = \int d^4x \int d^4\theta \bar{\Phi} \Phi$$

$$\text{s.t. } \bar{\Phi}(\gamma^\mu, \bar{\theta}) = \bar{\chi}(\gamma^\mu) + \bar{\theta} P_L \chi + \bar{\theta} \bar{\theta} F(\chi)$$

$$\text{and } D_\alpha \bar{\Phi} = 0$$

$\rightarrow \int d^4\theta \bar{\Phi} \Phi$ must be modified in such a way that the "F-term" yields S_F above.

* Assume $h_F = 4$ is the Chiral superfield to be the local superant version of the product multiplet.

$\rightarrow$ Obtain the superant version of the chiral mult whose

lowest weight is $\bar{\Phi}$

$$\delta F = [(h+2)\lambda_0 + i(h-3)\lambda_T]F + \frac{1}{\sqrt{2}}\bar{\epsilon}\not{D}P_L\chi + \delta\epsilon(1-h)\not{D}P_L\chi$$

Simplifies if $h=1$, removing the last term, adding as it π .

→ Then δF is similar to that of $\delta\bar{Z}$ at $h_{\bar{Z}}=2$

$$\delta\bar{Z} = [2\lambda_0 - i2\lambda_T]\bar{Z} + \frac{1}{\sqrt{2}}\bar{\epsilon}\not{D}P_L\chi'$$

$$\rightarrow \delta F|_{h=1} = [2\lambda_0 - 2i\lambda_T]F + \frac{1}{\sqrt{2}}\bar{\epsilon}\not{D}P_L\chi$$

• From the complex conjugate we identify

$$P_L\chi' = \not{D}P_R\chi$$

We can now use ^{the} Q-supersymmetry transform of $P_L\chi'$ to find the "top" F' component of the multiplet.

$$Z' = \bar{F}, \quad P_L\chi' = \not{D}P_R\chi, \quad F' = D^c\bar{Z}$$

In superspace this is simply $D_{\dot{\alpha}}D_{\dot{\beta}}\Phi^\dagger$

$$S_F = \int d^4x \int d^4\theta D^{\dot{\alpha}}D_{\dot{\alpha}}\Phi^\dagger$$

→ Superconformal multiplet w/ $h=2$

Recap: $\{(Z, P_L\chi, F), h_{\bar{Z}}=1\}$ i.e. Z is a scalar field

→ $\{(\bar{F}, \not{D}P_R\chi, D^c\bar{Z}), h=2\}$

→ The product of these two $\{(Z\bar{F}, \dots), h=3\}$

→ The superconf. mult. for $\Phi^\dagger$ must be realised differently.

→ components must be similar to $\delta F, \delta P_L\chi, \dots$

Recall

$$\square^c\phi := M^{\alpha\beta}\not{D}_\alpha\not{D}_\beta\phi, \quad D_a\phi := e_a^m(\partial_m - h b_m)\phi$$

to compute $D_a(D_b\phi)$ take $\delta(D_b\phi)$ and replace coeff. by their respective generators

$$D_a Z = e_a^m(\partial_m Z - h b_m Z - i h A_m^{(n)} Z - \frac{1}{\sqrt{2}}\bar{\psi}_m P_L\chi)$$

∴ To find $D^c Z$ we must take $\delta(D_a Z)$ fuck....

$$(1) \rightarrow \delta(e_a^m(\dots)) = e_a^m(\delta(\dots)) + e_a^m(\partial_m \delta Z - h \delta(b_m Z) - i h \delta(A_m^{(n)} Z) - \frac{1}{\sqrt{2}}\delta(\bar{\psi}_m P_L\chi))$$

$$\begin{aligned}
&= (\lambda_a^b e_b^u + \lambda_b e_a^u)(\epsilon) + e_a^u \left[(h\lambda_b + i h\lambda_T) \partial_m z + \frac{1}{\sqrt{2}} \bar{E} \partial_L \chi \right. \\
&\quad - h 2 \lambda_a^b e_{ub} z - h b_m (h\lambda_b + i h\lambda_T) z - h \frac{1}{\sqrt{2}} \bar{E} \partial_L \chi \\
&\quad \left. (-i h (h\lambda_b + i h\lambda_T) z - i h \frac{1}{\sqrt{2}} \bar{E} \partial_L \chi) A_m^{(R)} \right. \\
&\quad \left. + \frac{1}{\sqrt{2}} \bar{E} e_b^u \gamma_b \partial_L \chi - \frac{1}{\sqrt{2}} \bar{F}_m \left[(h + \frac{1}{2}) \lambda_b + (h - \frac{3}{2}) i \lambda_T \right] \partial_L \chi + \frac{1}{\sqrt{2}} \partial_L (Dz + F) \epsilon \right. \\
&\quad \left. + \sqrt{2} h z \partial_L \chi \right]
\end{aligned}$$

$$\begin{aligned}
&= \lambda_a^b D_b z + \lambda_b D_a z + e_a^u h \lambda_b \partial_m z + h b_m \lambda_b z - i \lambda_b z A_m^{(R)} \\
&\quad - \frac{1}{\sqrt{2}} \bar{F}_m \lambda_b \partial_L \chi
\end{aligned}$$

$$\Rightarrow \lambda_a^b D_b z + 2 \lambda_b D_a z + \dots$$

$$\boxed{D^c z = \eta^{ab} D_a D_b z = e^a{}^u (\partial_m D_a z - 2 b_m D_a z + \omega_{mab} D^b z + 2 f_{ma} z - i A_m^{(R)}) D_a z - \frac{1}{\sqrt{2}} \bar{F}_m \partial_L D_a \chi + \frac{1}{\sqrt{2}} \bar{F}_m \gamma_a \partial_L \chi}$$

* To check the Q-susy of $\partial_L \chi$, one needs the transformation of $\gamma^a D_a \partial_R \chi$. Since $\partial_L \chi' = \not{D} \partial_R \chi$

$$\Rightarrow \delta_Q \partial_L \chi' = \delta_Q (\not{D}^a D_a \partial_R \chi)$$

Note since there are no terms "underlined" in the expressions so $\delta(P_a) = 0$ and we simply have

$$\begin{aligned}
\delta_Q (\partial_L \chi') &= \gamma^a D_a (\not{D} (\frac{1}{\sqrt{2}}) (Dz + F) \epsilon) = \frac{1}{\sqrt{2}} \not{D} \partial_R (Dz + F) \epsilon \\
\not{D} \not{D} &= D^c + \underbrace{\gamma^{ab} D_a D_b}
\end{aligned}$$

$\rightarrow$ Ignore second term, as γ^{ab} is not in the final result of a susy transformation

$$\Rightarrow \delta_Q \chi = (D^c \bar{z} + \not{D} F) \epsilon$$

$$\Rightarrow F' = D^c z$$

The product of $(z, \partial_L \chi, F)$ and $(\bar{F}, \not{D} \partial_R \chi, D^c z)$

$$\boxed{
\begin{cases}
z(z\bar{F}) = z\bar{F} \\
\partial_L \chi(z\bar{F}) = \partial_L \chi \bar{F} + z \not{D} \partial_R \chi \\
F(z\bar{F}) = F\bar{F} + z D^c z - \bar{\chi} \partial_L \not{D} \chi
\end{cases}$$

* Last term relates to the S_F term in global susy.

* Scalars appear w/ the superconformal extension of the d'Alembertian
 We substitute these into $S_F = \int d^4x e \text{Re} \{ F\bar{F} + \frac{1}{2} \bar{\Psi}_\mu \gamma^\mu \not{D} \chi + \frac{1}{2} z \bar{\Psi}_\mu \gamma^{\mu\nu} \not{D} \psi_\nu \}$

$$S_{SC} = - \int d^4x e \text{Re} \{ F\bar{F} + z D^c z - \bar{\chi} \not{D} \chi + \frac{1}{2} \bar{\Psi}_\mu \gamma^\mu (\not{D}_L \chi \bar{F} + z \not{D}_R \chi) + \frac{1}{2} z \bar{F} \bar{\Psi}_\mu \gamma^{\mu\nu} \not{D}_R \psi_\nu \}$$

* Expanding $\not{D}$ shows that only the $F\bar{F}$ term will remain in the Action.

Gauge-fixing: Must Gauge all symm. which are not part of the Super-Poincaré: (U_1, D, T_R, S)

(i) SCT's: Since $\delta_\lambda^a(\chi_a) \psi_\mu = +2(\lambda_a)_\mu$ we can use $E[b_\mu = 0]$ λ_a to go to the gauge $\psi_\mu = 0$

(ii) Dilatations: We fix $|z|$ instead of $\phi = \frac{1}{K}$
 $\rightarrow$ Must also fix the T_R -symm as well.

(D-gauge & T-gauge): $z := \frac{\sqrt{3}}{K}$

(iii) S-gauge: $\chi = 0$

$$\Rightarrow D_a z = -i A_a z$$

$$D_a \not{D}_L \chi = e_a^\mu \not{D}_\mu \left[\frac{1}{2} (i K z - F) \psi_\mu - \sqrt{2} z \phi_\mu \right]$$

$$D^c z = -i e^{an} (\partial_n A_a z + \omega_{nab} A^b z) + 2 f_n^\mu z - A^a A_a z - \frac{1}{2} \bar{\Psi}^\mu \not{D}_\mu \chi$$

$$\therefore S_{SC} = - \int d^4x e \text{Re} \left\{ F\bar{F} + \frac{3}{K^2} \left(-i e^{an} (\partial_n A_a + \omega_{nab} A^b) + 2 f_n^\mu - A^a A_a \right) + \frac{1}{2} \bar{\Psi}_\mu \gamma^\mu z \not{D}_R \left[e_a^\mu \not{D}_\mu \left(\frac{1}{2} (i K z - F) \psi - \sqrt{2} z \phi \right) \right] \right\}$$

$\mathcal{O}(1)$ term

$$\Rightarrow S_{SC} = - \int d^4x e \left[F\bar{F} + \frac{3}{K^2} \left(2 f_n^\mu - A^a A_a - \frac{1}{2} \bar{\Psi}_\mu \gamma^{\mu\nu} \not{D}_\nu \psi + \frac{1}{2} i \bar{\Psi}_\mu \gamma^{\mu\nu} \not{D}_\nu \psi \right) \right]$$

using $f_n^\mu = -\frac{1}{2} R^n{}_\mu - \frac{1}{2} (R(\omega) - \bar{\Psi}_a \gamma^{\mu\nu} \not{D}_\nu \psi_b)$

$$\gamma^\mu \psi_\mu = -\frac{1}{2} \gamma^{\mu\nu} R'_{\mu\nu}(\omega)$$

$$\psi^a \psi_b = -\frac{1}{4} \gamma^{ab} \psi \psi$$

$\therefore$ Superconformal Gravity: in $D=4$

$$S_{SG} = \int d^4x \left[\frac{1}{2\kappa^2} (R(\omega(e, \psi)) - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} (\partial_\nu + \frac{1}{4} \omega_a{}^{bc}(e, \psi) \gamma_{bc}) \Psi_\rho + 6A^\mu A_\mu) - F\bar{F} \right]$$

* When the gravitino torsion terms are split from $\omega_a{}^{bc}(e, \psi)$ we find an extra 4-gravitino terms in $N=1, D=4$

supergravity

Recall: $D=4, N=1$ supergravity: $S = S_2 + S_{3/2}$

$$S_2 = \frac{1}{2\kappa^2} \int d^4x e e^a{}_\mu e^b{}_\nu R^{\mu\nu}{}_{ab}(\omega)$$

$$S_{3/2} = -\frac{1}{2\kappa^2} \int d^4x e \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu \Psi_\rho$$

$\Rightarrow$ First order formalism $\omega_a{}^{bc} = \omega_a{}^{bc}(e) + K_a{}^{bc}$

$$K_a{}^{bc} = -\frac{1}{2} (T_{[ab]c} - T_{[ca]b} + T_{[bc]a})$$

∇ solve for $T_{ab}{}^c$ and plug back into action

$$S = \frac{1}{2\kappa^2} \int d^4x (e e^a{}_\mu e^b{}_\nu R^{\mu\nu}{}_{ab} - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu(e, \psi) \Psi_\rho + \mathcal{L}_{\text{torsion}}^{(4)})$$

• F and A^μ are auxiliary fields.

$\rightarrow$ Include Supergrav. transformations: $\delta_\alpha e_\mu^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_\mu$

ψ_{max} : For the gravitino we must take the gauge fixing conditions into account: $b_\mu = 0, z = \frac{\sqrt{3}}{\kappa}, \chi = 0$

Note: $\chi=0$ (S-gauge) is not super symmetric and is only super-invariant for combinations of Q & S-susy.

$$\delta_P \chi = \left[(h^{-1/2}) \lambda_0 + i(h^{-3/2}) \lambda_1 \right] P_L \chi + \frac{1}{\sqrt{2}} P_L (\gamma^z F) \epsilon + \sqrt{2} h z P_L \eta$$

$$\rightarrow \text{under Gauge: } \delta_P \chi = \frac{1}{\sqrt{2}} P_L F \epsilon + \frac{\sqrt{6}}{\kappa} h P_L \eta + \frac{1}{\sqrt{2}} P_L \gamma^\mu (\partial_\mu z - i h A_\mu z)$$

$$\delta_P \chi = \frac{1}{\sqrt{2}} P_L F \epsilon - \frac{i}{\sqrt{2}} h P_L \chi \left(\frac{\sqrt{3}}{\kappa} \right) + \frac{\sqrt{6}}{\kappa} h P_L \eta \stackrel{!}{=} 0$$

$$h=1: P_L \eta = i \frac{1}{2} P_L \chi = \frac{\sqrt{3}}{2\sqrt{3}} F P_L \epsilon$$

why?

1

$$\therefore P_L \psi = \frac{1}{2} i \gamma_5 \psi, \quad \psi \rightarrow \frac{1}{\sqrt{2}} P_L \psi \quad (\text{Decomposition } \psi \rightarrow \psi_L)$$

$$\therefore \delta \psi_m \sim \gamma_a e_m^a \left(\frac{1}{2} i \gamma_5 \psi + \frac{1}{2\sqrt{3}} F \psi \right)$$

$$\delta \psi_m = -\frac{1}{4} \lambda^{\alpha\beta} \gamma_{\alpha\beta} \psi_m + \left(\partial_m + \frac{1}{4} \omega_m^{\alpha\beta} \gamma_{\alpha\beta} - \frac{3}{2} i A_m \gamma_5 \right) \psi + \frac{1}{2} \gamma_a e_m^a i \gamma_5 \psi + \frac{1}{2\sqrt{3}} \gamma_a e_m^a F \psi - \frac{1}{2} \lambda_0 \psi_m + \frac{3}{2} i \lambda_5 \gamma_5 \psi_m$$

$$\Rightarrow \delta P_L \psi_m = P_L \left(\partial_m + \frac{1}{4} \omega_m^{\alpha\beta} \gamma_{\alpha\beta} - \frac{3}{2} i A_m + \frac{1}{2} i \gamma_5 + \frac{1}{2\sqrt{3}} \gamma_m F \right) \psi$$

$$(\lambda_5 \equiv 0)$$

* Must find susy transformations for A_m and F :

$$\begin{aligned} \delta A_m &= \partial_m \lambda_5 - i \bar{\epsilon} \gamma_5 \psi_m + \frac{1}{2} i \bar{\epsilon} \gamma_5 \psi_m \\ &= -i \bar{\epsilon} \gamma_5 \psi_m + \frac{1}{2} i \bar{\epsilon} \gamma_5 \psi_m \end{aligned}$$

* We can use the lemma of cov. derivatives:

$\delta A_m \propto \psi_m$: Gauge $U(1)$ for superpoincaré supergravity
So do not consider this term:

→ Look at $\bar{\epsilon}$ term: Recall

$$\psi_m = -\frac{1}{2} \gamma^{\mu\nu} R_{\mu\nu}^A(Q) + \frac{2}{12} \gamma_m \gamma^{\alpha\beta} R_{\alpha\beta}^A(Q)$$

$$\Rightarrow \delta A_m = -i \bar{\epsilon} \gamma_5 \left(-\frac{1}{2} \gamma^{\mu\nu} \hat{R}_{\mu\nu}^A(Q) + \frac{2}{12} \gamma_m \gamma^{\alpha\beta} \hat{R}_{\alpha\beta}^A(Q) \right)$$

$$\text{where } \hat{R}_{\mu\nu}^A := \gamma_{\mu\nu}^A - 2 B_{[\mu}^B M_{\nu]}^A$$

$$\delta(\epsilon) B_{\mu}^A = \partial_{\mu} \epsilon^A + \epsilon^C B_{\mu}^B f_{BC}^A + \epsilon^B M_{\mu}^A$$

$$\gamma_{\mu\nu}^A = 2 \partial_{[\mu} B_{\nu]}^A + B_{\mu}^C B_{\nu}^D f_{CD}^A$$

$$R_{\mu\nu}^A(Q) = 2 \left(\partial_{[\mu} - \frac{3}{2} i A_{\mu]} \gamma_5 + \frac{1}{4} \omega_{\mu}^{\alpha\beta} \gamma_{\alpha\beta} \right) \psi_{\nu]}$$

$$P_L \hat{R}_{\mu\nu}^A(Q) = 2 P_L \left(\partial_{[\mu} + \frac{1}{4} \omega_{\mu}^{\alpha\beta} \gamma_{\alpha\beta} - \frac{3}{2} i A_{\mu} + \frac{1}{2} i \gamma_5 + \frac{1}{2\sqrt{3}} \gamma_{\mu} F \right) \psi_{\nu]}$$

$$\text{For } F: \delta F = \left[2 \lambda_0 F + \frac{1}{2} \bar{\epsilon} \gamma^{\mu} \left(-\frac{1}{2} (\not{D} \psi + F) \psi_m - \not{F} \psi_m \right) \right]$$

ψ_m can be omitted and $\lambda_0 = 0$

$$\text{Note } \gamma^{\mu} \psi_m = -\frac{1}{6} \gamma^{\mu\nu} \hat{R}_{\mu\nu}^A(Q)$$

$$\therefore \delta F = \frac{\sqrt{3}}{6} \bar{\epsilon} \not{F} \gamma^{\mu} \hat{R}_{\mu\nu}^A(Q)$$

with these symmetry transformations, the supergravity multiplet closes off-shell

* For any chiral theory, we can also add superpotential terms $\sim \int d^4\theta W(\Phi) \delta^2(\theta) + \int d^4\theta W^\dagger(\bar{\Phi}) \delta^2(\bar{\theta})$
 However, we must check that $\mathcal{P}$ is super conformal.

$$S_F = \int d^4x [F W'(z) - \frac{1}{2} \bar{F} P_L W''(z) \chi] \\
 \text{which, we can expand } W(z) = \sum_{n=-\infty}^{\infty} a_n z^n \\
 W'(z) = \sum_{n=-\infty}^{\infty} n a_n z^{n-1}, \quad W''(z) = \sum_{n=-\infty}^{\infty} n(n-1) a_n z^{n-2}$$

$$\delta F = (h+1) \lambda_D F$$

$$\delta S_F = \int d^4x (h+1) \lambda_D F W'(z) + \lambda_D F \sum_{n=-\infty}^{\infty} n a_n \delta(z^{n-1}) + \dots \\
 = \int d^4x (h+1) \lambda_D F W'(z) + \lambda_D F \sum_{n=-\infty}^{\infty} n(n-1) a_n z^{n-2} (\lambda_D h z) \\
 = \int d^4x (h+1) \lambda_D F W'(z) + \lambda_D F \sum_{n=-\infty}^{\infty} n(n-1) a_n z^{n-1} (\lambda_D h) \\
 = \int d^4x \sum_{n=-\infty}^{\infty} a_n \lambda_D F z^{n-1} [(h+1) + (n-1)h]$$

$$W \sim a z^3 \Rightarrow \delta W = \lambda_D 3 a h z^3$$

$$\Rightarrow h+1 + (n-1)h = 4 \quad \text{since } c: h_c = -4$$

$$h n + 1 = 4 \Rightarrow \boxed{n = 3}$$

$$\boxed{W(z) = C z^3} \quad \text{This has weight } \underline{h_w = 3}$$

$\Rightarrow S_F$ from above must have a chiral field with lowest weight $h_z = 3$. Thus, we must replace z w/ z^3 in S_F .

For $\chi = 0$,

$$W(z) = \int d^4x e^{i \beta C z^2 \text{Re}\{F\}}$$

$\rightarrow$ This is for Global supersymmetry. However, note that $\text{Re}\{F\} \mapsto e \text{Re}\{F + \frac{1}{2} \bar{\Psi}_m \gamma^m P_L \chi + \frac{1}{2} z \gamma^{\mu\nu} P_R \Psi_\nu\}$

Superpotential term, for (S-Gauge $\chi = 0$)

$$\int d^4x e^{i \beta C z^2 \text{Re}\{F + \frac{1}{2} z \gamma^{\mu\nu} P_R \Psi_\nu\}}$$

$$\text{Re}\{F_R\psi\} \Rightarrow \frac{1}{2}(1-\gamma_4)\psi \mapsto \frac{1}{2}\psi$$

$$\therefore S_c := \int d^4x e \left[3c z^2 \text{Re}\{F\} + \frac{3}{4} c z^3 \bar{\Psi}_\mu \gamma^{\mu\nu} \Psi_\nu \right]$$

* The full conformal theory is given by

$$S_{\text{CSG}} = S_{\text{SG}} + S_c$$

With S_c F has slightly more complex dynamics i.e. its E.O.M:

$$\text{Re}(F) = \frac{1}{2}(F + \bar{F})$$

$$S_{cc} \sim \int d^4x e(-F\bar{F})$$

$$\Rightarrow \frac{\delta S}{\delta \bar{F}} = -F + \frac{3}{2} c z^2 = 0$$

$$\therefore F = \frac{3}{2} c z^2 = \frac{9c}{2\kappa^2} \quad (\text{On-shell})$$

Plugging this back in we find

$$S_{\text{SG}} \sim -\left(\frac{18}{4} \frac{c^2}{\kappa^4}\right)$$

$$S_c = \int d^4x e \left(3c \left(\frac{3}{2\kappa^2}\right) \left(\frac{9c}{2\kappa^2}\right) + \frac{1}{4} c \frac{3^{3/2}}{\kappa^3} \bar{\Psi}_\mu \gamma^{\mu\nu} \Psi_\nu \right)$$

$$\therefore S_{\text{SG}} + S_c = \int d^4x e \left(\frac{1}{2\kappa^2} \left[R(\omega(e, \psi)) - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu(\omega(e, \psi)) \Psi_\rho + m_{3/2} \bar{\Psi}_\mu \gamma^{\mu\nu} \Psi_\nu \right] \right. \\ \left. - \left(\frac{31}{2} \frac{c^2}{\kappa^2}\right) + \frac{(3^{3/2})c^2}{\kappa^2} + \frac{1}{2} c \frac{3^{3/2}}{\kappa} \bar{\Psi}_\mu \gamma^{\mu\nu} \Psi_\nu \right]$$

$$S_{\text{SG}} + S_c = \int d^4x \frac{e}{2\kappa^2} \left[(R(\omega(e, \psi)) - \Lambda) - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu(\omega(e, \psi)) \Psi_\rho + m_{3/2} \bar{\Psi}_\mu \gamma^{\mu\nu} \Psi_\nu \right]$$

$$\Lambda := -\frac{81c^2}{2\kappa^2}$$

$$m_{3/2} := -\frac{2}{L} = \frac{2}{2} \kappa^2 c z^3 = \frac{3^{3/2} c}{2\kappa}$$

$$\Lambda = -6m_{3/2}^2$$

$\Rightarrow$ This is an Anti-de Sitter (AdS_4) super(conformal) gravity with parameter L . Thus the superpotential WZJ parameter c determines the value of the cosmological Constant Λ .

The S-susy contributes to the Q-susy of ψ_m

Claim: This leads to $\delta\psi_m = \hat{D}_m \epsilon$, where

$$\hat{D}_m \epsilon := (D_m - \frac{1}{2L} \gamma_m) \epsilon = (D_m + \frac{1}{4} \omega_{mab} \gamma^{ab} - \frac{1}{2L} \gamma_m) \epsilon.$$

Proof: We work on shell w/ $A_c = 0$, $\lambda_T = \lambda_D = 0$

$$\begin{aligned} \delta_Q \psi_m &= (D_m + \frac{1}{4} \omega_{mab} \gamma^{ab}) \epsilon - \gamma_m \left(-\frac{\kappa}{2\sqrt{3}} \left(\frac{\partial_c}{2\alpha^2} \right) \epsilon \right) \\ &= D_m \epsilon + \frac{3^{3/4} \kappa}{4} \gamma_m \epsilon = D_m \epsilon + \left(\frac{1}{2} (-\frac{1}{L}) \right) \gamma_m \epsilon \\ &= (D_m + \frac{1}{4} \omega_{mab} \gamma^{ab} - \frac{1}{2L} \gamma_m) \epsilon \quad \square \end{aligned}$$

Claim: The supersymmetry algebra of the preserved E-susy contains a Lorentz transformation w/ parameter

$$\lambda^{ab} = \left(\frac{1}{4L} \right) \bar{\epsilon}_2 \gamma^{ab} \epsilon_2$$

How?

The AdS_5 super algebra: $N=1$, $\{Q_a, Q_b\} = -\frac{1}{2} (\gamma^a)_{ab} P_a - \frac{1}{8L} (\gamma^a)_{ab}$

Result: $P_L \chi = \frac{1}{2} P_L \chi \epsilon - \frac{\kappa}{2\sqrt{3}} F P_L \epsilon$

$$\begin{aligned} [\delta_S(\chi), \delta_Q(\epsilon)] P_L \chi &= \delta_S \left(\frac{1}{2} P_L (\chi \epsilon + F) \epsilon + \sqrt{2} h \chi P_L \chi \right) \\ &\quad - \delta_Q (\sqrt{2} h \chi P_L \chi) \\ &= \delta_S \left(\frac{1}{2} P_L F \epsilon + \sqrt{2} \chi \left(\frac{1}{2} \right) \right) = -\sqrt{2} h \left(\frac{-\kappa}{2\sqrt{3}} \right) \delta_S \rightarrow 0 \end{aligned}$$

→ Come back to this.

the 1990s, the number of people in the world who are under 15 years of age is expected to increase from 1.2 billion to 1.5 billion.

As the world's population grows, the demand for food and other resources will increase. This will put pressure on the environment and on the world's food supply.

One way to meet this demand is to increase the amount of food that is produced. This can be done by using more land for agriculture or by increasing the productivity of the land that is already being used.

Another way to meet this demand is to reduce the amount of food that is wasted. This can be done by improving the way that food is stored and distributed.

There are many other ways to meet the world's growing demand for food and other resources. It is up to us to decide which way is the best.

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the fact that the *Journal of the American Medical Association* (JAMA) is the only journal in the field to have a dedicated section for the publication of letters to the editor.

There are a number of reasons why the JAMA is the only journal in the field to have a dedicated section for the publication of letters to the editor. One reason is that the JAMA is the only journal in the field that is published by a medical association.

Another reason is that the JAMA is the only journal in the field that has a dedicated section for the publication of letters to the editor.

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